Log Book Entry 1.1

NB2420 Electronic Instrumentation · Manual 1.1 · Voltage divider

Hand in via BrightSpace

Name(s): | Partner:

Date: | Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

PREDICT, SIMULATE, MEASURE

Tasks A1, A2, S1, I1, I2, connected to C1 and C2

Supply is 5 V. Fill the predicted columns as homework, the simulated column after LTspice, and the measured columns at the bench. Leave a cell blank rather than guessing, then say why in the analysis.

R_1	R_2	U_{out} pred. A1	I pred. A2	U_{out} sim. S1	U_{out} meas. I1	I meas. I2
100 Ω	100 Ω					
1 kΩ	1 kΩ					
100 kΩ	100 kΩ					
10 MΩ	10 MΩ					
1 kΩ	220 Ω					

Combined dividers

Tasks A4, S2, I3, I4 and I5, leading to C3

I3: record each separate divider output relative to the common ground.

R_3, R_4 output (V): | R_1, R_2 output (V):

Configuration	U_{out} predicted	U_{out} simulated	U_{out} measured
Two dividers, outputs connected (I4)			
Stacked dividers, left circuit (I5)			
Stacked dividers, right circuit (I5)			

Setup *Sketch the circuit you built and label the nodes you probed, or name a photo you hand in beside this entry.*

Photo or file reference:

Function generator and loading Tasks A3, A5,A6, I6, I7, C1 to C4

C4: sine, 2 V RMS displayed, 1 kHz, zero offset in every row. Enter 1 kΩ in the blank load cell. Disconnected means one resistor-bank lead is removed; the DMM stays connected. No selector jumper gives 1 MΩ.

Physical load	Generator load setting	U_{rms} expected (V)	U_{rms} measured (V)	Explanation
Disconnected	HIGH Z			
	HIGH Z			
50 Ω	HIGH Z			
50 Ω	50 Ω			
Disconnected	50 Ω			

Graph *Plot measured U_{out} or I against resistance. Label both axes with a unit.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: